Gradient – Diagnostics – Static Test and Fidelity Diagnostic

Equipment Room

Main Disconnect Panel — CB3, TB1-15-16, CB2

Remote MRU

O_2 Monitor Control Unit — Meter, Ctrl Board, Alarm, Light

Customer-Supplied Outlet

Site E-Net

Facility Power Filters

Facility Utilities — Facility Coolant Supply, Facility Coolant Return

E-Off Switch

Operator Workspace

Temp/Hum Sensor

Server (Option)

Service Laptop

Optional Speaker

GOC

- IRDC (Option) — J5, J4, J3, J2, J1
- USB-Serial Converter
- Ethernet Switch — 1, 2
- GOCAA — J12, J14, J17, J18, J15, J7, J6
- Host PC — Ethernet B, Lwr Graphic Card Port 1, RS232, USB, USB, USB, USB, Upr Graphic Card Port 1, Ethernet A, Line In, Line Out
- J10, J4, J6, J9, J2, GOC, J5, PDU, J1, J3

Global Modem (Option)

Key-board

Mouse

Primary Display

Pneumatic Patient Alert System

PGR

- PDM — J29, J25, L1/L2/L3, J27 J28 J24
- VRE 2 (32-ch) — J5, Net 3, Net 2, Net Mgmt, J1
- VRE 1 — J1, Net Mgmt, Net 2, Net 3, J5
- ETH 1, 3, 2, 4 Term Server
- Aux Board — J4, J6, J1, J2/J3, J9, J11, J10
- XRFD LS1-A, LS1-B, LS2-A, LS2-B, LS3-A, LS3-B, LS4-A, Term — PDU Top 1, PDU Top 1, PDU Front
- J4 XGD Fan x3
- NB Amp — CB2, J14, J9, J10, J20, J5, J11, J7, J15, J19 J16 J8 J12 J6, J4, J3, J18, J1, J2
- 16 Port E-net Sw — 3, 2, 1, 4, 5, 6, 15, 12, 11, 14, 13, 7
- CB1, J1, J4, J3, Recon, RF/MNS, J5, J2, PEN, PGR PDU, Host, J6/J7/J8, J11/J12/J13
- Unused
- J3 J6, J2, J102, J19 J18 J10 J9 J17, J14 J15, J13 J12, J4
- J8 NB AIF, UPM1 — Det, J1 J2 J4 J5 J3, Proc J3
- UPM2 — J1 Det, J2 J4 J5 J3, Proc J3
- CAM — J2 ETH, SCP2 PMC, J3, 10/100 Base, AGP2, SRF/TRF, STIF J4 J3, J18/J19, J23 I/O Panel, IRF3 J10 J8, J14 J13 J12 J11
- Cronus Chassis — XPS CB (J13/J16, J12/J15, J11/J14), XGA CB (J13/J16, J12/J15, J11/J14), J10/J11 J6 J9 GMC J8 J7, J2 J1, P1, J3 J4 J5, Cronus Chassis Optical Fiber Communications
- Pwr XPS x3 — J2 J10, J1 J11, J8, J9 J3, J4/J5, J4/J5
- XGA x3 — J8 J7 J16, J13 J15, J14 J5, J12 J6 J17 J18, J1/J2, J1/J2, J1/J2
- Manifold Supply — SMA Coax 9-pin, LEM Icoil x3 — Grad Pulse In, Grad Pulse Out
- Term Block x3
- Manifold Return
- J20, J204, J201, J200, J203, J202, J205, J22, J11, J103, J8

Diagnostic Data Path: Static Test Fidelity Diagnostic

The Static Test and Fidelity Diagnostic data paths originate in:
1. The Cronus Chassis GMC board and pass
2. To the XGAs, via the Cronus backplane.

Door Switch

Coolant Removal Kit

Data Path – General

The Host PC is connected to the Gradient chain through the Ethernet Switch. The Gradient information, in turn, passes through the Ethernet Switch to the CAM Chassis via the AGP Board. From the AGP board, information is passed to the SRF/TRF board via the CAM backplane. The SRF/TRF board operates as the brains of the process, passing the signal on to the other two boards via the CAM backplane.

Once fully processed the signal is sent to the Cronus Chassis in the XGD before the Gradient Pulse is generated.

- F/O Wvgde
- Temp/Hum Sensor
- Exciter — J19, J20, J21, J29, J32, J30, J15 J14 J17 J1
- Distribution — J4 J2 J3
- CFB — J18, J19, J7/J8, J1/J2, J5/J6, J3/J4, J17
- Driver Module — J14 J8 J19 J17, J11, J12, J13, J2, J20, J1, J7, J6 J23 J25 J21
- DD Filter — J94 J96 J92, J74 J76 J72
- PHPS — J12 J9, J37/J38, J2, J43, J39, J54, J41, J11, J1, J5, J10, J40, J42, J35, J36, J44, J8
- SRPS — J6, J5, J1, J51, J52, J46, J48, J45, J50, J47, J3
- J17, J4, J3, J9, J5, J6, J16
- J39 J54 J41 J40 J42 J35 J36 J51 J52 J46 J48 J45 J50 J47 J29 J32 J30 J34 F11 F10 F9 F8 F7 F6 F5 J31 J33
- Scan Room Blower Box
- Patient Comfort
- Hub

- Mag Mon — J5, J10, J8, J9, J7
- HEC — HEC Power Box (Cryo, Power Supply, GC, PE, Blower); Signal Box (J24, J11 J12); Grad HtXchr; Grad Res; Grad Pump; Flow Meter; Filter; Blower HtXchgr; PGR Pump; PGR Res; PGR HtXchr; Grad/RF Coil Blower
- JR Power Cryo-Cooler Compressor JC — Water Return, Water Supply, Gas Return, Gas Supply

- SPW — J115, J116, J117, J16, J106, J108, J107, Term Conn F1 F2 F3 F4, X Y Z Gradient Filter X Y Z, J128, J121 J122 Xmit Sw1 J120 J123, J104 Wvgde, J105 Wvgde
- J118 J17
- J100 Wvgde, J101 Wvgde, J111 Wvgde, J110 Wvgde, Patient Alert Wvgde
- Primary MRU
- E-Off Switch
- O_2 Monitor — Sensor, Alarm, Light
- Mag Enclosure Chimney
- Gradient Bus Bar — X, Y, Z
- Mag I/O — J1, J3, J6, J4, J2, J5, J14, J15, J12, J11, J13, J8, J7, J9, J10, J17, J16
- FO I/F — J28, J20, J21

Scan Room

- Mag Bore — J6 J7, Body Coil J2 J3 J9, Bore-PAL L R, Left Arc Cover, Ambient Temp Snsr, Bore Temp Sensors, Front Mic, Airflow Sensors, XRMB X Y Z Gradient Temp Sensor, J101 MAG-Grad Coil, J100 MAG-Grad Coil, J104 Grad Coil Air, Patient Air
- Magnet — J3 P403 Terminal Box, Pressure Transducer, Reed Switch, J2 RUO PreAmp J1, Coldhead Pwr Sleeve, Coldhead Supply Return, Airflow Sensor Module, J9/J10 J2 PAC, SRI4 — J23, J10, J34, J3, J20, J32, J24, J17, J1 J4, J5, J11/J12, J13/J14, J33, J21, J31, J2, J22
- PreAmp & Combiner — J11, J10, J9, J1 J5, J2 J6, J3, BHYB J7 J4, LCP J11 J3, RCP J11 J3, IRD J1
- J11 J2, J9 TDM J1, J10, J12
- J9 RRx1, J7, J1, J4/J5 J8 J2
- J7 RRx2 (32-ch), J1, J4/J5 J2

- Rear Pedestal — J14 J13 Dummy Load J12HYB; PED Antenna; Lights
  - RF Hub — Hub Backplane (bias, power, status/control); J102J101, J104, J103
  - PDB — J1 J3, J2, J4, J5, J6
  - MCDB
  - RFCB — J9 J13 J12, J16, J17, J10, J14/J15, J7, J6 J1 J2, J8, J3 J4; Debug Port
  - RFSB — J1–J6 (ch 17-32), (ch 1-16) J1–J6
  - J4 J6, J3 TIF, J1 J2 J5
  - Long Drive ASM (MTR) J1 J2
  - POS Enc
  - J3 EOT Sensor, HOME Sensor, EGRESS Sensor
  - LPCA — J4 J2, Port A J5 J1, Port P2 J4, Port P1 J5, Rear Mic
  - Load A
  - Load B
  - Reroute Box — J10 J7, P2A P3A, P2B P3B, P1A P4A, P1B P4B, J8 J9
  - J1 PED I/O J2
  - Rear Pedestal Air

MAG-PAC J4

PAC I/F

Squeeze Bulb

Speaker

Left E-Stop Button

Rt E-Stop Button

Patient Table
- Coil Port P3
- Coil Port P4
- Touch N Go_R
- Touch N Go_L
- Table Hydraulics

Dock
- J2 Mag
- J1 Dock
- J3 Left
- ODU

Key

TBD = To Be Determined (Future Enhancement)

Cabinet · Chassis/ Sub-system · Part

Data Cables · Coolant · Power · Pulse · Data Paths

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